

RESEARCH ARTICLE

## Discovery of Kv3.1 channel inhibitors reveals VU0521426 as a state-dependent inactivator preferentially active against pathogenic gain-of-function mutants

 Rahul U. Chandrappa,<sup>1\*</sup> Vaishali Satpute Janve,<sup>2\*</sup> Emily L. Days,<sup>3</sup> and  Jerod S. Denton<sup>1,2</sup>

<sup>1</sup>Department of Pharmacology, Vanderbilt University, Nashville, Tennessee, United States; <sup>2</sup>Department of Anesthesiology, Vanderbilt University Medical Center, Nashville, Tennessee, United States; and <sup>3</sup>High Throughput Screening Core Facility, Vanderbilt Institute of Chemical Biology, Vanderbilt University, Nashville, Tennessee, United States

### Abstract

Kv3.1 voltage-gated potassium channels play a critical role in regulating neuronal excitability, and dysregulation driven by gain-of-function (GoF) mutations has been implicated in neurological disease. Although Kv3.1 potentiators have been at the forefront of drug development as a means to enhance neuronal firing, progress towards small-molecule Kv3.1 inhibitors has been limited. Here, we address this gap with the discovery of novel and structurally diverse Kv3.1 channel inhibitors identified through a high-throughput screening of over 50,000 compounds. Among these, VU426 emerged as the most potent compound with an IC<sub>50</sub> of 4.3  $\mu$ M. VU426 induces pronounced, state-dependent inhibition of outward K<sup>+</sup> current with sustained depolarization, indicating stabilization of an inactivated channel conformation accessed from the open state. Functional characterization of four GoF mutations (V425M, M430I, V432M, and V434L) in the S6 pore lining domain demonstrated that VU426 exhibits 1.5- to 5-fold enhanced potency toward pathogenic GoF Kv3.1 mutants relative to wild-type channels. Automated patch clamp electrophysiological studies revealed that V432M and V434L mutations embedded deep in the S6 domain had the greatest sensitivity to VU426. Despite its potency, VU426 exhibited limited selectivity for Kv3.1 over related Kv channels. Together, these findings identify novel Kv3.1 inhibitors and highlight a pharmacological strategy for targeting clinically identified pathogenic variants.

**NEW & NOTEWORTHY** This study identifies the first structurally diverse collection of small-molecule inhibitors of Kv3.1 potassium channels, highlighting VU426 as a moderately potent, state-dependent inactivator. Notably, VU426 displays enhanced potency against multiple pathogenic gain-of-function Kv3.1 mutants, revealing a mechanism-based strategy for selectively targeting disease-associated channel variants.

*automated patch clamp; gain-of-function; HTS; KCNC1 mutations; Kv3.1*

### INTRODUCTION

Voltage-gated potassium (Kv) channels of the Kv3 family are uniquely adapted to support high-frequency neuronal firing due to their rapid activation and deactivation kinetics and relatively depolarized voltage-dependent activation. Among these, Kv3.1 (encoded by *KCNC1*) channels are fundamental drivers of the high-frequency action potential and neuronal oscillations (1–3). Rapid Kv3.1 channel activation facilitates fast neuronal membrane repolarization after action potential initiation, whereas fast deactivation reduces afterhyperpolarization, enabling a subsequent action potential to occur in less than a millisecond (4–6). Kv3.1 channels are abundantly expressed in fast-spiking GABAergic parvalbumin-expressing interneurons (PV-INs) of the neocortex, hippocampus, and basal ganglia (7). Enriched expression of Kv3.1 channels in PV-INs makes them critical regulators of the excitatory/inhibitory balance of neuronal circuits and potent drug targets (8–10).

Pathogenic missense *KCNC1* variants produce a broad phenotypic spectrum, including epilepsy (11, 12), intellectual disability (13, 14), developmental encephalopathies (15), schizophrenia (16), and autism (17). Functional studies indicate that most clinically reported *KCNC1* mutations result in loss of function; however, a small subset of missense mutations produce gain-of-function (GoF) effects that paradoxically also lead to developmental encephalopathies and even epilepsy (11, 14). These GoF variants often accelerate transitions into open states at hyperpolarized membrane potentials where the wild-type (WT) channels are normally closed (11, 14). This premature Kv3.1 channel opening can shunt action potentials and incapacitate PV interneurons to maintain their rapid firing patterns, leading to an imbalance in excitation and inhibition (18). This is also considered the primary mechanism of the aforementioned *KCNC1*-related channelopathies. This emerging clinical landscape underscores a clear therapeutic gap: while the field has focused primarily on Kv3 activation as a strategy to rescue loss-of-



\*R. U. Chandrappa and V. Satpute Janve contributed equally to this work.  
Correspondence: J. S. Denton (jerod.s.denton@vumc.org).  
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function states, there remains virtually no pharmacological framework for counteracting GoF-driven hyperactivity in Kv3.1.

Tetraethylammonium (TEA) and 4-aminopyridine have been used historically (2, 19–22), but these agents lack the specificity and mechanistic precision required to probe Kv3 gating or to serve as viable therapeutic leads. Selective small-molecule Kv3.1 inhibitors have not been developed or characterized in detail, leaving a significant unmet need for tools capable of modulating Kv3.1 gating in a state-dependent manner. This shortage of inhibitors also limits basic neuroscience research. PV interneuron circuit depends on tight control of the fast repolarization and high-firing frequency of action potentials, yet researchers currently lack small molecules that can selectively slow Kv3-mediated repolarization without engaging numerous other  $K^+$  channels. A selective Kv3.1 inhibitor would enable bidirectional modulation of fast-spiking properties, provide a tool for testing mechanistic structure-function hypotheses, and support modeling of PV interneuron dysfunction in disease-relevant contexts. Here, we report the discovery and characterization of several novel and structurally diverse inhibitors of Kv3.1 and describe defined state-dependent and mechanistic properties of the most potent inhibitor, termed VU0521426 (VU426). Surprisingly, we find that VU426 was more efficacious on GoF mutations identified in patients with developmental delay/intellectual disability or epileptic encephalopathy.

## METHODS

### Cell Culture

Cells were cultured in Dulbecco's Modified Eagle Medium (Gibco, 11965-092) supplemented with Tet-tested and heat-inactivated fetal bovine serum (10%; R&D Systems, S10350H), penicillin-streptomycin (2 mM; Gibco, 15140-122), Blasticidin HCl (10 mg/mL; Gibco, A11139-03), and Hygromycin B (50 mg/mL; Invitrogen, Carlsbad, CA, 10687-010) at 37°C in a humidified atmosphere containing 95% air/5% CO<sub>2</sub>.

### Generation of Mutant Kv3.1 Plasmid Vectors

The pcDNA5/TO-hKv3.1b plasmid vector encoding the human *KCNK1* isoform variant 2 (NM\_004976.4) was synthesized by GenScript. Site-directed mutagenesis was used to generate Kv3.1 mutants using high-fidelity PCR (Phusion, Thermo Fisher Scientific) with custom oligonucleotides to generate point mutations (Sigma-Genosys) using standard protocols. Parental DNA strands were digested with DpnI (New England Biolabs, R0176S) following PCR amplification of mutated primers, followed by bacterial transformation into NEB 10-beta competent *Escherichia coli* cells (New England Biolabs, Cat. No. C3019H). Transformed cells were cultured at 37°C for 1 h with shaking (200 rpm). Transformed cells were plated on LB agar plates with ampicillin selection antibiotic and incubated overnight at 37°C. Four isolated colonies were selected and cultured in LB broth supplemented with ampicillin for 16–18 h at 37°C with shaking (200 rpm). Plasmid DNA was purified using Qiagen Plasmid Plus Mini Kit (Cat. No. 12943) according to the manufacturer's protocol, and the mutated pcDNA was verified by Sanger sequencing. For large-scale amplification, correctly sequenced clones were grown in

30 mL LB broth with ampicillin and purified using Qiagen Plasmid Plus Midi Kit (Cat. No. 12943).

### Cell Line Generation

Monoclonal cell lines expressing wild-type (WT) Kv3.1 and polyclonal cell lines expressing mutant Kv3.1b(V425M), Kv3.1b(M430I), Kv3.1b(V432M), or Kv3.1b(V434I) channels were generated using T-Rex-HEK293 cells (Thermo Fisher Scientific) with tetracycline-regulated expression. T-Rex-HEK293 cells were transfected with pcDNA5/TO-hKv3.1b plasmid vectors using Lipofectamine LTX PLUS reagent (Thermo Fisher Scientific, Cat. No. A12621) following the manufacturer's instructions. Transfected cells were then selected for Kv3.1b expression by subjecting antibiotics selection 48 h after transfection (Blasticidin HCl 5 µg/mL and Hygromycin B 250 µg/mL). Cells viable after antibiotic selection were amplified to establish stable cell lines. Channel expression was induced using 1 µg/mL tetracycline 24 h before all experiments. Cell lines were tested for Kv3.1 channel expression using thallium assay and/or manual patch clamp recordings.

### Quantitative Kv3.1 Thallium Assay

Stable T-Rex-HEK293 cells expressing WT and mutant Kv3.1 channels were plated in cell culture flasks (75 cm<sup>2</sup>; Thermo Fisher Scientific) and cultured overnight in the culture media as previously described. Approximately 16 h before assay initiation, Kv3.1 expression was induced by treating cells with 1 µg/mL tetracycline. On the day of the assay, cells were rinsed with assay buffer (Hanks' balanced salt solution supplemented with 20 mM HEPES) and incubated for 1 h at room temperature in the dark with Brilliant Thallous AM dye (Ion Bioscience, San Marcos, TX, Cat. No. 1381), a thallium-sensitive fluorescent indicator. Excess dye was removed by washing the cells with assay buffer. Cells were then plated into poly-L-lysine-coated, black-walled, clear-bottom 384-well plates (BD Biosciences, Bedford, MA) at a density of 20,000 cells per well in 20 µL of culture medium. Fluorescence imaging was performed using a Panoptic Kinetic Imaging Plate Reader (WaveFront Biosciences, Franklin, TN). Baseline fluorescence measurements were acquired at 1 Hz (482 nm excitation/536 nm emission) before compound treatment. Compounds were prepared at 2× assay concentration and added to each well, 20 µL/well, followed by 4-min incubation. Subsequently, 10 µL of thallium-containing stimulus buffer containing 12.5 mM Ti<sub>2</sub>SO<sub>4</sub> (50 mM; ION Bioscience, 7040S), 1 mM MgSO<sub>4</sub>, 1.8 mM CaSO<sub>4</sub>·H<sub>2</sub>O, 5 mM D-glucose, 10 mM HEPES, and 125 mM NaHCO<sub>3</sub> was simultaneously added to each well of the plate with a robotic liquid handler. Fluorescence measurements were then acquired for 2 min to monitor thallium influx.

### Compound Discovery

A high-throughput thallium flux assay of 50,391 small molecules from the Vanderbilt Institute of Chemical Biology (VICB) Discovery Collection was conducted against the stable T-Rex-HEK293 cells expressing WT Kv3.1 channels to screen for novel activators and inhibitors of the channel. Compounds were tested in a single replicate at a final concentration of 10 µM. Fluorescence data were collected using Panoptic software and the Waveguide system (WaveFront Biosciences), and then imported into custom analysis software developed at VICB.

Fluorescence intensity over time for each well was normalized to its baseline value ( $F/F_0$ ). The change in fluorescence was calculated between 0 and 22 s after thallium addition and expressed as a percentage relative to the response induced by a dimethyl sulfoxide (DMSO) control on each 384-well plate. Compounds were considered hits if they showed no prethallium activity (e.g., absence of fluorescent artifacts) and met both  $Z$ -score and robust  $Z$ -score thresholds, defined as at least three standard deviations from the mean and three absolute median deviations from the median. Hit compounds were retested and confirmed to have no activity against HEK293T cells before being tested in eight-point concentration-response curves (CRCs) ranging between 30 nM and 100  $\mu$ M. The rate of change in fluorescence was calculated between 0 and 6 s after thallium addition and expressed as the relative percent inhibition response between DMSO control (0%) and a 10 mM TEA (100%) on each 384-well plate. CRCs were plotted using GraphPad Prism (GraphPad Software, San Diego, CA) and fitted with a four-parameter logistic model to calculate  $IC_{50}$  values for ranking compound potency.

### Automated Patch Clamp Electrophysiology

Kv3.1 channel-mediated  $K^+$  currents were recorded using the SyncroPatch 384 automated patch clamp (APC) instrument (Nanion Technologies) in the perforated patch configuration as previously described (23). Cell lines stably expressing WT Kv3.1 and mutant Kv3.1 channels were dissociated right before the experiment. Cells were dissociated shortly before the experiment and suspended at a density of  $1\text{--}2 \times 10^6$  cells/mL in ice-cold standard external physiological solution containing (in mM) 140 NaCl, 4 KCl, 2  $CaCl_2$ , 1  $MgCl_2$ , 5 glucose, and 10 HEPES (pH 7.4). Then cells were placed in a cell reservoir in the instrument at  $10^\circ\text{C}$  with stirring at 200 rpm. A potassium fluoride (KF)-based internal solution with 0.83  $\mu$ M escin (MilliporeSigma, E1378) was applied to the bottom of the NPC-384 chip (four-hole, medium-resistance, Nanion Technologies, 08 3007), followed by the addition of 20  $\mu$ L of a divalent-free solution and then 20  $\mu$ L of the cell suspension in each well using an automated pipette. A catch pressure of  $\sim 200$  mBar captured the cells on the chip, and 8 mM Ca-enriched  $N$ -methyl- $D$ -glucamine 60 (NMDG60) solution was added to improve seal resistance. Kv3.1 currents were evoked from a holding potential of  $-80$  mV by stepping the membrane potential in 10-mV increments from  $-100$  to  $+80$  mV, then returning to  $-80$  mV, repeating every 20 s (Supplemental Fig. S1A). Baseline currents were measured for 3 min, followed by a 5-min exposure to Kv3.1 inhibitor VU413, with a final 3-min addition of 12.5 mM tetraethylammonium (TEA) to conclude the experiment. Leak currents were estimated from subthreshold voltage steps and subtracted offline from all recorded currents before analysis. The intracellular recording solution contained (in mM) 100 KF, 10 KCl, 10 NaCl, and 10 EGTA (pH 7.2). The reference NMDG60 solution contained (in mM) 140 NMDG- $Cl^-$ , 80 NaCl, 4 KCl, 2  $CaCl_2$ , 1 mL  $MgCl_2$ , 5 glucose, and 10 HEPES (pH 7.4). The divalent-free solution was  $Ca^{2+}$ - and  $Mg^{2+}$ -free NMDG60 solution. The seal enhancer solution was prepared by adding 8 mM  $CaCl_2$  to the NMDG60 solution.

APC data were analyzed using the DataControl 384 software (Nanion Technologies). Before estimating leak currents, each well was inspected to meet the quality control (QC) criteria of

less than  $-1$  nA of holding current at  $-80$  mV, stable seal resistance, and drug responses throughout the recording. Only the wells that met the QC criteria were included. Percent inhibition was normalized to steady-state currents at  $+40$  mV measured from a full block application of 12.5 mM TEA (Supplemental Fig. S2). Peak currents were recorded as the maximum amplitude within 200 ms following the start of the voltage step, which captures peak currents at all voltage steps. Steady-state currents were measured for current-voltage ( $I$ - $V$ ) and  $IC_{50}$  values calculations and were calculated as the mean amplitudes in 100 ms before the end of the voltage step.  $IC_{50}$  values were determined by fitting the Hill equation to current versus concentration-response curves (CRCs) using variable-slope nonlinear regression analyses using GraphPad Prism version 10.6.1 (GraphPad Software). For CRCs, the mean steady state currents at  $+40$  mV were used. For conductance-voltage ( $G$ - $V$ ) curves, conductance ( $G = I/V - V_K$ ) values were calculated based on the potassium reversal potential of  $V_K$  of  $-86.50$  mV, based on the composition of internal and external solutions.  $G$ - $V$  curves were fitted with the Boltzmann sigmoidal function using GraphPad Prism version 10.6.1. Selectivity of VU426 was tested using eight-point concentration response curves on Kv1.1, Kv2.1, Kv3.2, and Kv3.3 (Supplemental Fig. S2). The Kv1.1, Kv2.1, Kv3.2, and Kv3.3 channels were stably expressed in tetracycline-induced HEK293T polyclonal cell lines, similar to Kv3.1 cell lines used throughout the experiments conducted in these studies. Voltage step protocols were similar to all other experiments (Supplemental Fig. S1).

### Chemicals

Powder of VU0521426 (F5822-2236) and all of the top 20 inhibitor hits from the screen were purchased from Life Chemicals. See the Vanderbilt University IDs and their corresponding external Life Chemical IDs in Fig. 1E. TEA was purchased from Sigma-Aldrich (121-44-8).

### Kv3.1 Molecular Modeling

Rendering of the Kv3.1 (PDB: 7PQT) channel protein showing the location of the residues altered due to GoF mutations was generated using UCSF Chimera.

### Data Analysis

Statistical analysis for the primary screen was performed using the Vanderbilt high-throughput screening (HTS) custom software Waveguide. Additional curve-fitting analyses were performed using GraphPad Prism. Repeated-measures one-way ANOVA followed by Dunnett's post hoc multiple-comparison testing was used to compare compound-treated conditions to the appropriate control within the same cell. Data are presented as means  $\pm$  SE. Individual data points represent recordings from individual cells, and the number of cells analyzed is indicated in the figure legends. Statistical significance was defined as  $P < 0.05$ .

## RESULTS

### Discovery of Structurally Diverse Kv3.1 Channel Inhibitors with HTS

We conducted a high-throughput screening (HTS) of 50,391 compounds to identify novel Kv3.1 channel inhibitors

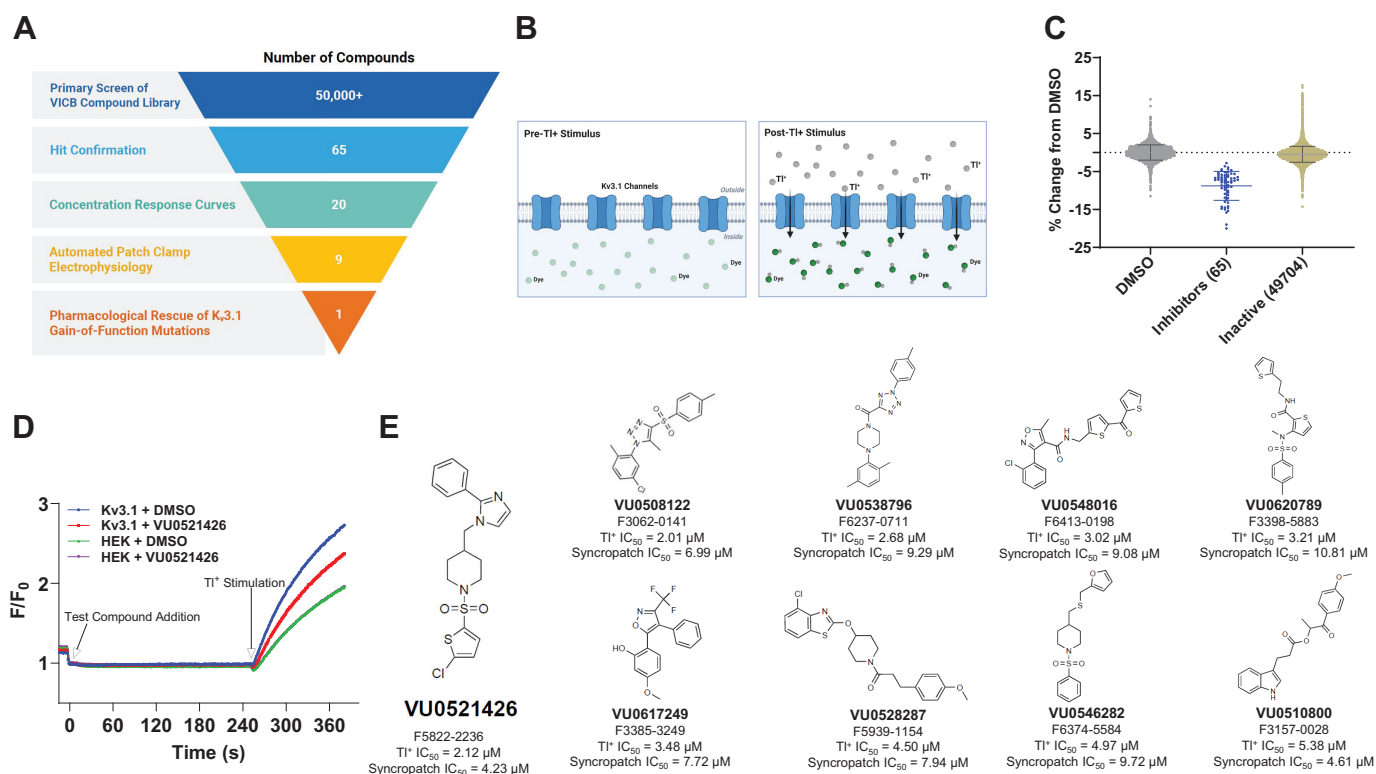


using a fluorescence-based thallium flux assay. The assay employed T-Rex-HEK293 cells stably overexpressing Kv3.1, in which channel opening permits thallium permeation and elicits an increase in intracellular fluorescence of the Thallo dye (Fig. 1B). This screening strategy identified 65 preliminary inhibitor hits that reduced thallium-induced fluorescence by  $\geq 3$  standard deviations below the mean response (*Z*-scores) of DMSO controls (Fig. 1C). Representative thallium flux traces showed that VU426 markedly reduced stimulus-evoked fluorescence in Kv3.1-expressing cells but had no effect in parental HEK293T cells, confirming channel-dependent activity (Fig. 1D). Following hit confirmation on parental HEK293T cells and concentration-response curve (CRC) prioritization in thallium assays, nine compounds were selected for further evaluation using automated patch clamp (APC) electrophysiology. The chemical structures and  $IC_{50}$  values obtained from thallium flux and APC assays for eight of these compounds are shown in Fig. 1E. From this

subset, we selected the most potent compound, VU0521426 (VU426), a sulfonamide-containing molecule (Fig. 1E, left), as our lead inhibitor for further investigation.

### VU426 Induces Kv3.1 Channel Inactivation

To validate the thallium assay results and elucidate the mechanism of VU426-mediated inhibition, we employed APC electrophysiology, a robust approach for characterizing voltage-dependent gating and mechanisms of action of Kv channel modulators. Voltage-clamp recordings using a perforated patch configuration revealed large, outwardly rectifying Kv3.1 currents that increased in amplitude with progressive membrane depolarization. Channels were activated by 500-ms voltage steps from  $-100$  to  $+80$  mV in 10-mV increments from a holding potential of  $-80$  mV (Supplemental Fig. S1A). Under control conditions, Kv3.1 currents exhibited large sustained outward currents with



**Figure 1.** Discovery of VU0521426 in a high-throughput screen of 50,391 compounds. **A:** schematic overview of the screening funnel used to identify novel Kv3.1 inhibitors. A primary thallium flux screen of  $\sim 50,000$  small molecules from the Vanderbilt Institute of Chemical Biology (VICB) Discovery Collection yielded 65 preliminary hits, of which 20 were advanced to concentration-response curves (CRCs). Nine compounds progressed to automated patch clamp electrophysiology, and one lead inhibitor was evaluated for pharmacological rescue of clinically reported Kv3.1 (*KCNK1*) gain-of-function (GoF) variants. **B:** diagram depicting thallium flux assay used for HTS, hit confirmation, and concentration response curves for compound prioritization. Before stimulus with thallium buffer (top), intracellular dye remains quenched in the absence of thallium. Following the addition of external thallium (bottom), Kv3.1 channels permit thallium influx, increasing intracellular fluorescence. Inhibitors reduce thallium entry, decreasing signal amplitude. **C:** violin plots show summary of screening results. Wells were treated with either DMSO (negative control; 0% inhibition) or 10  $\mu$ M of test compound. Each point represents the value of a single well of a 384-well plate change in amplitude by subtracting maximum minus minimum values collected following thallium addition. Compounds that reduced thallium fluorescence  $\geq 3$  SD or MAD below DMSO response were defined as inhibitors, and 65 such compounds were identified. **D:** representative thallium flux time course traces for Kv3.1 expressing cells (Kv3.1 + DMSO, Kv3.1 + VU0521426) and parental HEK-293T cells (HEK + DMSO, HEK + VU0521426). Test compound was added at  $t = 0$  s for  $\sim 4$  min, followed by thallium stimulus to induce ion influx (arrow). **E:** chemical structures of the lead inhibitor, VU0521426 (VU426), and eight other structurally distinct, small molecule inhibitors identified in the primary high-throughput screen and advanced for automated patch clamp characterization against wild-type (WT) Kv3.1 channels are shown. For each compound, half-maximal inhibitory concentrations ( $IC_{50}$  values) determined from concentration-response curves (CRCs) using thallium flux assay ( $n = 4$  wells at each dose) and SyncroPatch automated whole cell electrophysiology ( $n \geq 3$  at each dose) are indicated. DMSO, dimethyl sulfoxide; HTS, high-throughput screening.

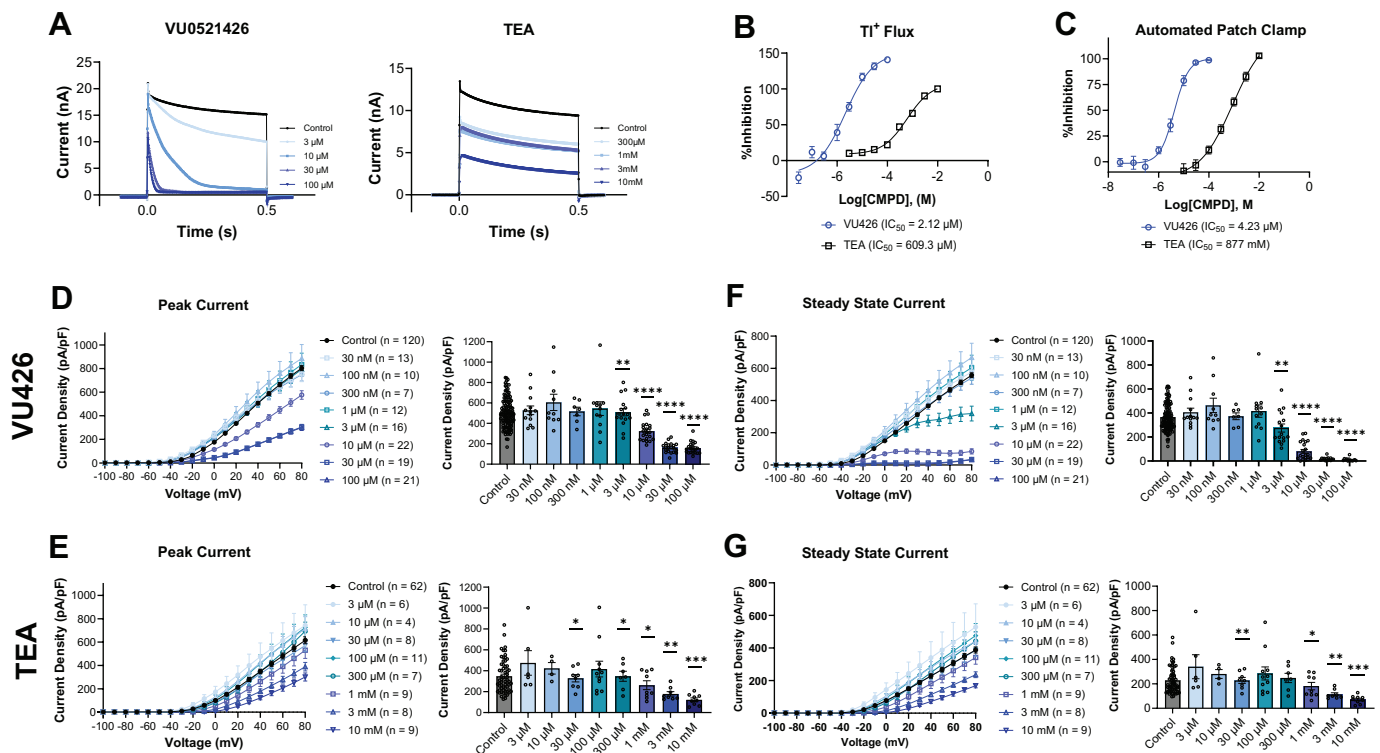
minimal intrinsic inactivation over the duration of the depolarizing step.

VU426 produced a concentration-dependent reduction in both peak and steady-state outward currents, accompanied by the emergence of rapid current decay within the first 50–100 ms of depolarization, consistent with pharmacologically induced inactivation of Kv3.1 channels (Fig. 2A, left). Tetraethylammonium (TEA), a well-established Kv channel pore blocker, was included as a reference control for channel inhibition. In contrast to VU426, although increasing concentrations of TEA reduced peak and steady-state currents, TEA failed to induce appreciable current inactivation during the 500-ms depolarizing step and did not fully suppress sustained Kv3.1 currents, even at saturating concentrations (Fig. 2A, right). Our data are in agreement with previous reports, indicating that TEA-mediated inhibition acts predominantly as a

pore blocker without significantly altering Kv3.1 gating, in contrast to the inactivation-stabilizing mechanism of VU426.

Concentration-response analysis of steady-state currents measured at +40 mV, a physiologically relevant voltage approximating the peak membrane potential during action potentials, revealed that VU426 inhibited Kv3.1 currents with an  $IC_{50}$  of 4.23  $\mu$ M (Fig. 2C, blue), whereas TEA displayed markedly weaker potency ( $IC_{50}$  = 877.0  $\mu$ M; Fig. 2C, black). The substantially greater potency of VU426 relative to TEA is consistent with results from the  $Tl^+$  flux assay (Fig. 2B; VU426  $IC_{50}$  = 2.12  $\mu$ M, blue; TEA  $IC_{50}$  = 609.3  $\mu$ M, black) and further supports the conclusion that VU426 acts as a state-dependent, gating-modifying inhibitor rather than a simple pore blocker.

To gain additional insight into the mode of VU426-mediated inhibition, we further analyzed APC recordings. Consistent



**Figure 2.** Electrophysiological characterization of VU426 inhibition of WT Kv3.1 channel currents using APC. **A:** representative current traces at +40 mV recorded during 500-ms voltage step protocols from −80 mV using automated patch clamp (APC) recording in a perforated patch configuration. Currents were recorded in the absence (control, black) or presence of increasing concentrations of the inhibitor VU426 (left; 3, 10, 30, and 100  $\mu$ M) and classical Kv channel blocker tetraethylammonium (TEA) (right; 0.3, 1, 3, 10 mM). Both inhibitors demonstrate a concentration-dependent reduction in outward  $K^+$  currents. VU426 resulted in a pronounced current inactivation and inhibited steady-state currents. **B:** concentration-response thallium flux inhibition curves for VU426 compared with the benchmark blocker TEA (10 mM defining 100% block). VU426 exhibits potent inhibition of Kv3.1 channels ( $IC_{50}$  = 2.12  $\mu$ M,  $n$  = 4 at each dose), whereas TEA shows weaker activity ( $IC_{50}$  = 609.3  $\mu$ M,  $n$  = 32 at each dose). **C:** concentration-response curves for inhibition of steady-state whole cell currents by VU426 (blue  $\circ$ ) and TEA (black  $\blacksquare$ ). Percentage inhibition was normalized to steady-state currents measured from a full block application of 12.5 mM TEA. Data were fit with a sigmoidal dose-response curve in prism to calculate half-maximal inhibitory concentrations ( $IC_{50}$ ), yielding an  $IC_{50}$  of 4.228  $\mu$ M for VU426 ( $n$   $\geq$  7 at each dose, means  $\pm$  SE) and 877.0 mM for TEA ( $n$   $\geq$  7 at each dose, means  $\pm$  SE). **D** and **E** (left): current-voltage ( $I$ - $V$ ) relationships for peak currents measured in the presence of increasing concentrations of VU426 (0.03–100  $\mu$ M) and TEA (0.003–10 mM). Curves represent means  $\pm$  SE, with the number of cells analyzed indicated in the legend. **Right:** bar graphs of peak current densities at +40 mV demonstrate a concentration-dependent inhibition of peak current with significant inhibition at high micromolar doses of VU426. Similar percentage inhibition is observed at hundreds of micromolar and low millimolar doses of TEA (repeated measures ANOVA with Dunnett post hoc multiple comparison testing to control;  $P$  values are indicated by asterisks above groups;  $*P$  = 0.05;  $**P$  > 0.01;  $***P$  > 0.001;  $****P$  > 0.0001). **F** and **G** (left):  $I$ - $V$  relationships for steady state currents recorded in the presence of increasing concentrations of VU426 (0.03–100  $\mu$ M) and TEA (0.003–10 mM), showing robust suppression of sustained Kv3.1 mediated currents. **Right:** bar graphs of steady state current densities at +40 mV reveal significant inhibition at low micromolar concentrations of VU426. In addition, 10–100  $\mu$ M doses of VU426 are able to abolish all Kv3.1 current during a 500-ms depolarizing step to +40 mV, unlike higher doses of TEA (repeated measures ANOVA with Dunnett post hoc multiple comparison testing to control;  $P$  values are indicated by asterisks above groups;  $*P$  = 0.05;  $**P$  > 0.01;  $***P$  > 0.001;  $****P$  > 0.0001). WT, wild type.

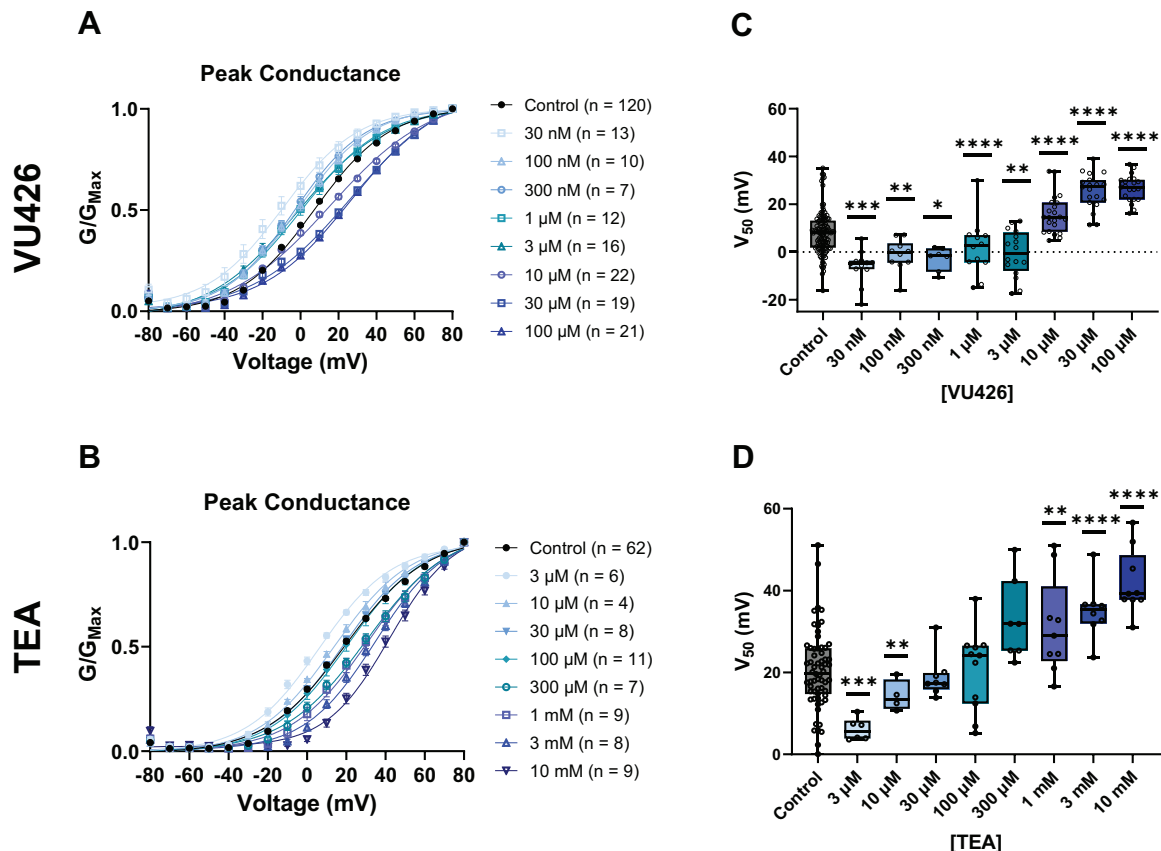
with representative current traces (Fig. 2A, left), current-voltage ( $I$ - $V$ ) relationships and current densities at +40 mV demonstrated that micromolar concentrations of VU426 (3–100  $\mu$ M) significantly reduced peak Kv3.1 current amplitudes, with the greatest suppression observed at depolarized membrane potentials (Fig. 2D). In contrast, TEA produced a significant reduction in peak current density only at concentrations  $\geq 300$   $\mu$ M and elicited a small but significant increase in current at 30  $\mu$ M (Fig. 2E).

Steady-state currents were markedly more sensitive to VU426 than peak currents, with near-complete suppression observed at concentrations of 10, 30, and 100  $\mu$ M (Fig. 2F, left). Repeated-measures analysis confirmed a significant reduction in steady-state current density at concentrations  $\geq 3$   $\mu$ M (Fig. 2F, right). For TEA, steady-state  $I$ - $V$  relationships showed dose-dependent reductions in current density at 1, 3, and 10 mM (Fig. 2F), mirroring its effects on peak currents. Notably, VU426-mediated inhibition was substantially greater

for steady-state currents compared with peak currents, particularly at higher depolarized potentials. This supports a mechanism in which VU426 preferentially inhibits open Kv3.1 channels by stabilizing an inactivated, nonconducting state rather than by simple pore block.

### VU426 Alters Voltage-Dependent Gating of Kv3.1 Channels

To determine whether VU426 modulates channel gating in addition to reducing current amplitude, we analyzed conductance-voltage ( $G$ - $V$ ) relationships derived from peak currents at each voltage step. Under control conditions, Kv3.1 channels exhibited the expected depolarization-activated conductance profile. Application of VU426 at 30 and 100  $\mu$ M produced a significant rightward shift in the  $G$ - $V$  relationship (Fig. 3A), consistent with reduced channel availability at depolarized voltages. This effect was quantified by measuring the voltage at half-maximal activation ( $V_{50}$ ), which was



**Figure 3.** Concentration-dependent modulation of Kv3.1 voltage dependence by VU426 compared with TEA. A and B: normalized conductance-voltage ( $G$ - $V$ ) relationships measured from Kv3.1 peak currents under control conditions and in the presence of increasing concentrations of VU426 (A) or the classical Kv channel inhibitor TEA (B). Peak conductance values were normalized to the maximal conductance ( $G/G_{\text{max}}$ ) for each condition and plotted as a function of membrane potential using a 500-ms depolarizing voltage step protocol using APC (Supplemental Fig. S2). Solid lines represent Boltzmann fits to the data using prism, and individual points indicate means  $\pm$  SE. The number of cells recorded for each condition is shown in the legends. Both VU426 and TEA produced depolarizing shifts in the voltage dependence of activation, with progressively steeper rightward shifts observed at only higher concentrations. However, both inhibitors produced a leftward shift at their respective low doses. C and D: summary box-and-whisker plots of the half-maximal activation voltage ( $V_{50}$ ) values derived from Boltzmann fits to peak  $G$ - $V$  curves from A and B, respectively, for VU426 (C) and TEA (D). Boxes represent the interquartile range with the median indicated by the central line, whiskers denote the minimum and maximum values, and individual points represent single cells. Statistical comparisons vs. control were performed using repeated measures one-way ANOVA followed by Dunnett post hoc multiple comparisons to the appropriate control, with adjusted  $P$  values indicated by asterisks above groups (\* $P$  = 0.05; \*\* $P$  > 0.01; \*\*\* $P$  > 0.001; \*\*\*\* $P$  > 0.0001). VU426 produced significant depolarizing shifts in  $V_{50}$  beginning at low micromolar concentrations (30 and 100  $\mu$ M), whereas TEA required higher concentrations to achieve comparable shifts (1–10 mM), consistent with potencies of the two inhibitors. APC, automated patch clamp; TEA, tetraethylammonium.

significantly depolarized at these concentrations (Fig. 3C). In contrast, 10  $\mu$ M VU426 did not significantly alter  $V_{50}$  relative to control, whereas lower concentrations ( $<10$   $\mu$ M) produced small but statistically significant hyperpolarizing shifts in voltage dependence of activation, indicating modest channel activation at more negative potentials.

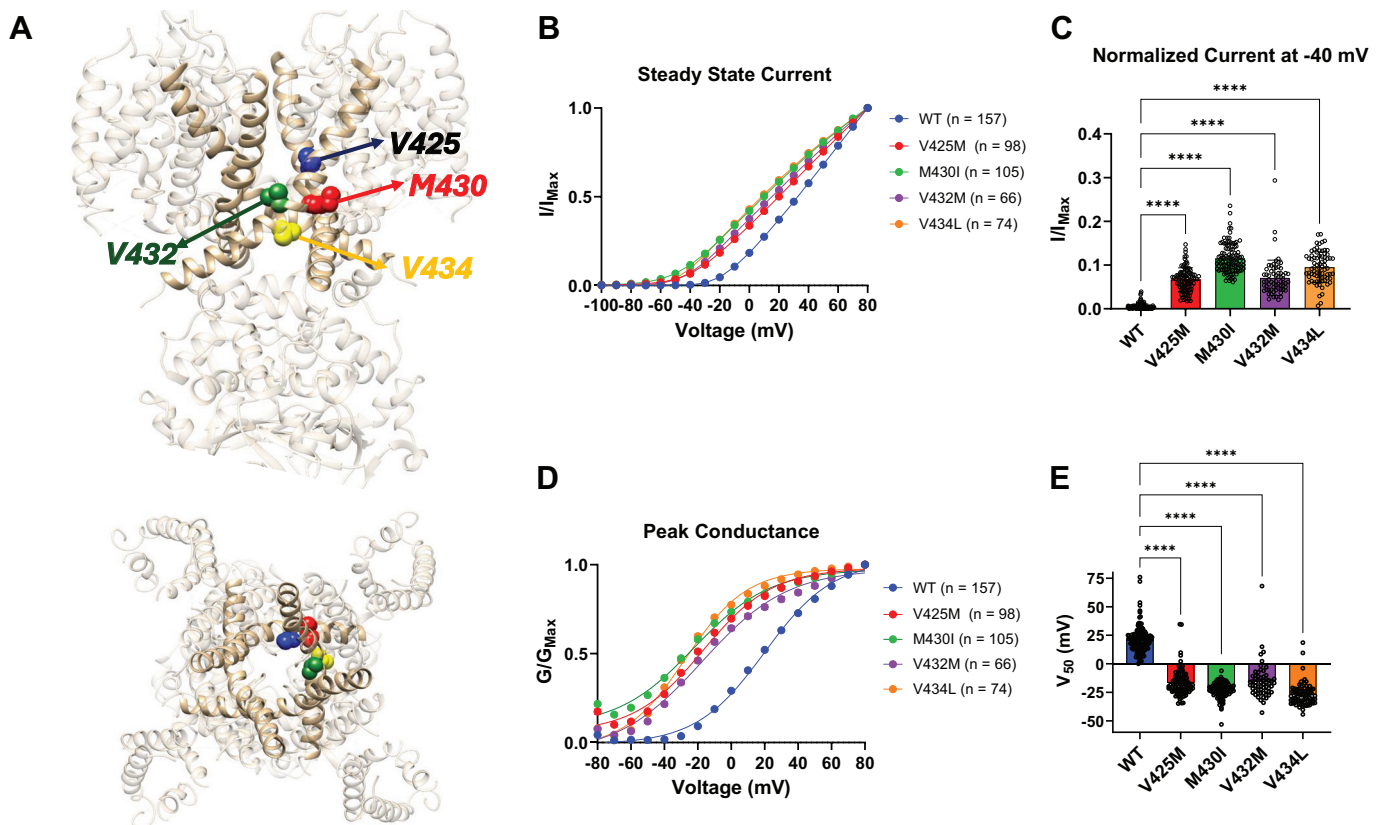
TEA produced qualitatively similar but less pronounced effects on peak  $G$ - $V$  relationships (Fig. 3B). Low micromolar concentrations of TEA (i.e., 3 and 10  $\mu$ M) significantly shifted  $V_{50}$  toward more hyperpolarized potentials, whereas only high millimolar concentrations (1, 3, and 10 mM) induced significant depolarizing shifts in  $V_{50}$  (Fig. 3D). The bidirectional effects of both inhibitors across concentration ranges suggest complex, state-dependent modulation of Kv3.1 voltage-dependent gating, with VU426 producing stronger gating perturbations consistent with its inactivation-stabilizing mechanism.

### Clinically Reported KCNC1 GoF Mutations Exhibit Hyperpolarized Channel Activation

Clinically reported *KCNC1* mutations V425M, M430I, V432M, and V434L confer a gain-of-function (GoF) phenotype

characterized by increased channel conductance and enhanced activation at hyperpolarized membrane potentials at which wild-type (WT) Kv3.1 channels are normally silent (11, 14). Structural mapping of these variants onto the cryo-EM structure of the Kv3.1 tetramer (PDB: 7PQT) revealed that all four mutations cluster within the S6 transmembrane segment, a pore-lining region that plays a critical role in gating and ion conduction (Fig. 4A). Top-down views of the structure show that residues V425, V432, and V434 project toward the central pore, whereas M430 is oriented toward the outward-facing surface of the S6 helix (Fig. 4A, right).

To assess whether automated patch-clamp (APC) electrophysiology could reliably capture the functional consequences of clinically relevant *KCNC1* mutations, we examined these previously characterized GoF variants (11, 14). Consistent with prior reports, APC recordings revealed a clear GoF phenotype for all four S6 mutations (Fig. 4, B–E). Normalized steady-state current-voltage relationships showed pronounced leftward shifts relative to WT channels, reflecting increased current densities across membrane potentials (Fig. 4B). Strikingly, mutant channels exhibited aberrant current flow at negative



**Figure 4.** Location of clinically reported *KCNC1* GoF mutations that exhibit hyperpolarized channel activation. **A:** location of clinically reported *KCNC1* GoF mutants mapped onto the apo cryo-EM structure of Kv3.1 (PDB: 7PQT) with the pore domains, consisting of the S5 and S6 regions, highlighted and the rest of the domains left transparent. GoF mutations cluster near the S6 region of the channel's pore domain. **B:** normalized steady-state current densities from recording of stable polyclonal cell lines expressing the respective GoF mutation reveal more hyperpolarized channel opening for all mutations relative to WT. The number of cells recorded for each mutant is shown in the legends. **C:** summary of steady-state activation from normalized steady-state current density at -40 mV confirms significant hyperpolarizing shifts in mutant channels. **D:** peak ( $G$ - $V$ ) relationships demonstrate left-shifted activation for all mutations compared with WT. The number of cells recorded for each mutant is shown in the legends. **E:** quantification of peak  $G$ - $V$   $V_{50}$  values shows consistent gain-of-function (GoF) across variants. Data are represented as means  $\pm$  SE; individual points represent individual wells. Statistics for pairwise comparisons between mutants and WT were conducted using one-way ANOVAs with Dunnett post hoc comparison to WT (\* $P$  = 0.05; \*\* $P$  > 0.01; \*\*\* $P$  > 0.001; \*\*\*\* $P$  > 0.0001). WT, wild type.



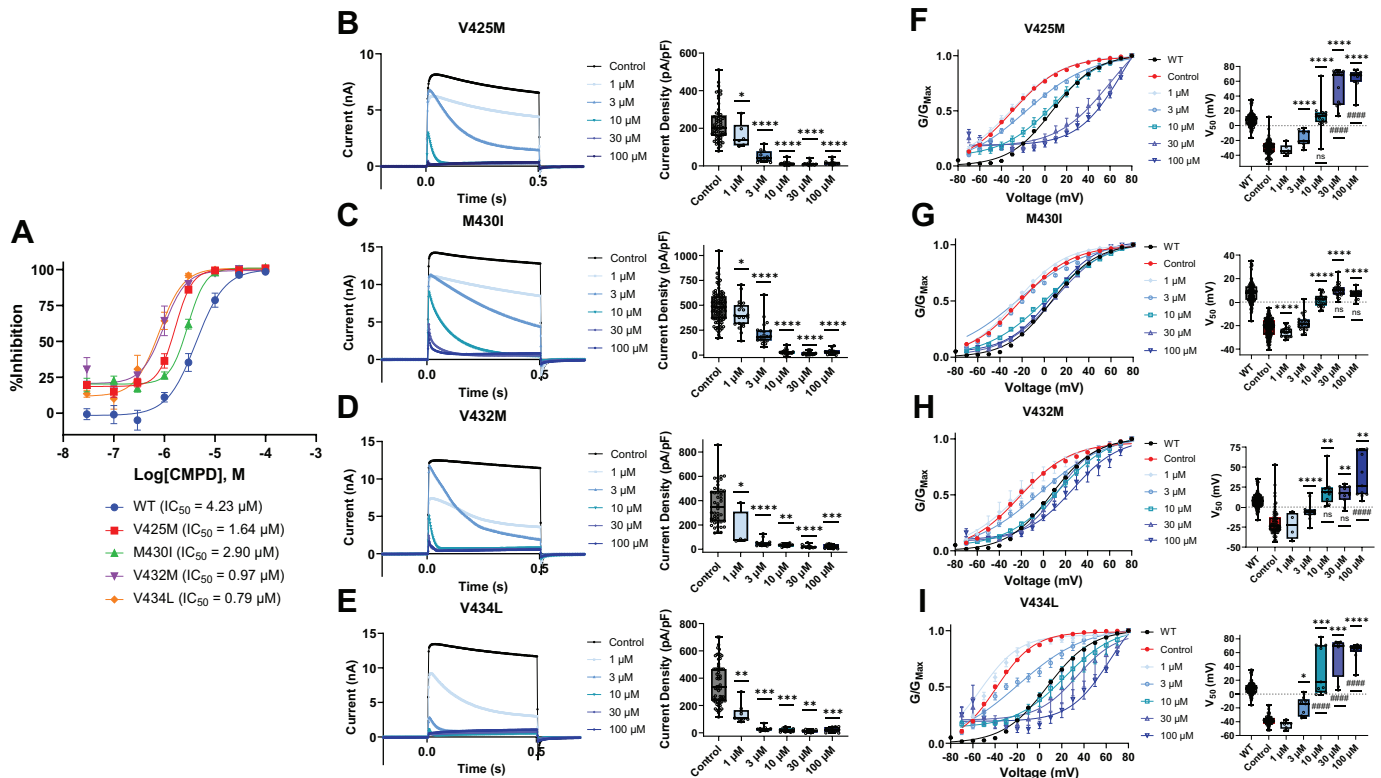
voltages where WT Kv3.1 channels remained quiescent. Normalized steady-state currents at  $-40$  mV confirmed significant current amplitudes for V425M, M430I, V432M, and V434L, in contrast to the negligible current observed for WT channels (Fig. 4C).

To determine whether these effects arose from altered voltage-dependent activation, we analyzed their  $G$ - $V$  relationships. All four GoF mutants exhibited significant leftward shifts in their  $G$ - $V$  curves (Fig. 4D), accompanied by hyperpolarizing shifts in the voltage of half-maximal activation  $V_{50}$  derived from Boltzmann fits (Fig. 4E). Together, these findings indicate that missense mutations within the S6 domain likely destabilize the closed state of Kv3.1 channels indicated by increased channel conductance at hyperpolarized potentials, which aligns with structure-guided hypotheses from Kv3.1 cryo-EM studies (24). Importantly, these results demonstrate that APC electrophysiology reliably recapitulates known gating defects in disease-associated mutations and provides an efficient platform for

functional interrogation of the expanding spectrum of pathogenic *KCNK1* variants (25).

### VU426 Exhibited Enhanced Potency Against Kv3.1 Gain-of-Function Mutants

After establishing APC electrophysiology as a platform for functional interrogation of Kv3.1 channel variants, we next evaluated whether VU426 could serve as a proof-of-concept tool inhibitor for mutation-targeted channel modulation. CRC analysis revealed that VU426 was  $\sim 1.5$ - to 5-fold more potent on all GoF Kv3.1 channels compared with the WT channels (WT  $IC_{50} = 4.228 \mu M$ ; V425M  $IC_{50} = 1.639 \mu M$ ; M430I  $IC_{50} = 2.898 \mu M$ ; V432M  $IC_{50} = 970.5$  nM; V434L  $IC_{50} = 788.5$  nM; Fig. 5A). Notably, the V432M and V434L variants, located deepest within the S6 pore-lining S6 region, exhibited the greatest sensitivity, with sub-micromolar  $IC_{50}$  values. Similar to WT Kv3.1 channels, application of VU426 induced pronounced current inactivation in all GoF mutants, with substantial reductions in both peak and steady-state currents



**Figure 5.** VU426 exhibits greater potency on Kv3.1 channels harboring GoF mutations in the S6 domain. **A:** concentration-response relationships for inhibition of steady-state current by VU426 in wild-type (WT) Kv3.1 and clinically reported gain-of-function (GoF) variants V425M ( $n \geq 5$  cells at each dose), M430I ( $n \geq 13$  cells at each dose), V432M ( $n \geq 4$  cells at each dose), and V434L ( $n \geq 3$  cells at each dose) stably expressed in polyclonal cell lines. Percentage inhibition was normalized to the maximally inhibited steady-state currents with  $12.5$  mM TEA. Data were fit with a sigmoidal dose-response curve in prism;  $IC_{50}$  values are indicated in the legend. **B–E:** representative whole cell current traces recorded from cells expressing V425M (B), M430I (C), V432M (D), or V434L (E) Kv3.1 channels at a depolarizing 500-ms step to  $+40$  mV from  $-80$  mV under control conditions and in the presence of increasing concentrations of VU426. Plots to the right show box-and-whisker plots of steady-state current densities at  $+40$  mV and reveal significant inhibition at low micromolar concentrations (repeated measures ANOVA with Dunnett post hoc multiple-comparison testing to control;  $P$  values for pairwise comparisons are listed above). **F–I:** peak conductance-voltage ( $G$ - $V$ ) relationships for V425M (F), M430I (G), V432M (H), and V434L (I) channels in control and VU426-treated conditions. Conductance was normalized to the maximal value ( $G/G_{max}$ ) and fit with a Boltzmann function to derive activation midpoints ( $V_{50}$ ). Plots to the right show box-and-whisker plots showing VU426-dependent shifts in  $V_{50}$  relative to control for each mutant channel (repeated measures ANOVA with Dunnett post hoc multiple-comparison testing to mutant control;  $P$  values are indicated by asterisks placed above bars;  $*P = 0.05$ ;  $**P > 0.01$ ;  $***P > 0.001$ ;  $****P > 0.0001$ ). Data for all curves in A–I are presented as means  $\pm$  SE. Additionally  $V_{50}$  values of WT cells in control condition were compared with mutant cells receiving varying doses of VU426 (one-way ANOVA with Dunnett post hoc multiple-comparison to WT values;  $P$  values for groups showing no significant (ns) difference or significantly higher  $V_{50}$  values to WT are indicated by hash marks (#) and placed below bars;  $\#P = 0.05$ ;  $\##P > 0.01$ ;  $\###P > 0.001$ ;  $\####P > 0.0001$ ). TEA, tetraethylammonium.



during depolarizing voltage steps (Fig. 5, B–E). Significant suppression of steady-state currents was observed at low micromolar concentrations ( $\geq 1 \mu\text{M}$ ) across all variants, consistent with enhanced compound sensitivity.

Analyses of  $G$ - $V$  relationships demonstrated that VU426 produced concentration-dependent depolarizing shifts in channel activation for all GoF mutants (Fig. 5, F–I), partially reversing their pathological hyperpolarized gating phenotype. Consistent with WT channels, VU426 induced significant depolarizing shifts in  $V_{50}$  relative to control conditions (Fig. 5F; see figure legend for values). At higher concentrations (i.e.,  $\geq 10 \mu\text{M}$ ), mutant  $V_{50}$  values approached those observed for WT channels (WT:  $7.93 \pm 0.81 \text{ mV}$ ;  $10 \mu\text{M}$  VU426-V425M:  $12.08 \pm 4.76 \text{ mV}$ ;  $30 \mu\text{M}$  VU426-M430I:  $10.09 \pm 1.13 \text{ mV}$ ;  $10 \mu\text{M}$  VU426-V432M:  $20.15 \pm 6.16 \text{ mV}$ ;  $10 \mu\text{M}$  VU426-V434L:  $36.28 \pm 10.49 \text{ mV}$ ). In contrast, lower concentrations of VU426 ( $\leq 1 \mu\text{M}$ ) produced modest but significant hyperpolarizing shifts in  $V_{50}$  relative to control, a phenomenon also observed for WT channels exposed to VU426 and TEA (Fig. 2, A–D), suggesting complex concentration-dependent effects on channel gating.

Collectively, these findings demonstrate that VU426 exhibits enhanced potency toward all tested GoF Kv3.1 variants and produces concentration-dependent depolarizing shifts in channel activation that partially normalize aberrant gating. These data establish VU426 as a potent gating modulator capable of functionally rescuing biophysical defects associated with S6 GoF mutations. Moreover, the increased sensitivity of these variants supports the hypothesis that S6 mutations enhance the ability of VU426 to stabilize a nonconducting, inactivated channel state, a mechanism that will require future analyses to quantify effects on inactivation rates and recovery.

### VU426 Inhibits Other Kv Channels

To assess the subtype selectivity of VU426 across voltage-gated potassium (Kv) channels, we measured concentration-dependent inhibition of steady-state currents from a panel of heterologously expressed Kv channels using automated patch clamp (APC) electrophysiology (Supplemental Fig. S2). VU426 inhibited multiple Kv channel subtypes, including Kv1.1, Kv2.1, Kv3.1, Kv3.2, and Kv3.3, with broadly similar efficacy. Nonlinear regression of concentration-response relationships revealed overlapping  $\text{IC}_{50}$  values across the tested channels, with Kv3.1 inhibition occurring at comparable efficacy and approximately two- to threefold lower apparent potency than several other Kv subtypes.

## DISCUSSION

In this study, we leveraged a high-throughput thallium flux screening strategy paired with APC electrophysiology to identify and mechanistically characterize small-molecule inhibitors of Kv3.1 channels. This approach led to the discovery of VU0521426 (VU426), a sulfonamide-containing compound that suppresses Kv3.1 activity through a state-dependent mechanism consistent with stabilization of nonconducting channel states. Beyond establishing VU426 as a moderately potent and informative tool compound, our findings demonstrate the utility of HTS platforms for uncovering previously unexplored modes of Kv3.1 inhibition and provide

a framework for targeting disease-associated pathological Kv3.1 channels.

Kv3.1 channels play a specialized role in enabling high-frequency firing in fast-spiking neurons, yet pharmacological tools to modulate their activity, particularly to suppress gain-of-function (GoF) variants, have been limited. Here, we implemented an HTS of over 50,000 compounds using a fluorescence-based  $\text{Tl}^+$  flux assay optimized for Kv3.1 activity, identified chemically diverse inhibitors that were subsequently prioritized by APC profiling (Fig. 1). This pipeline enabled rapid triaging of hits based not only on potency but also on kinetic and state-dependent properties that would be difficult to discern from fluorescence assays alone. The identification of multiple Kv3.1 inhibitors with micromolar potency (Fig. 1E), including VU426 as the most potent representative, highlights the feasibility of applying HTS strategies to voltage-gated potassium channels traditionally considered challenging drug targets.

A central finding of this work is that VU426 inhibits Kv3.1 channels through a mechanism distinct from classical pore blockade. Unlike TEA, which reduces current primarily by occluding the conduction pathway, VU426 produces pronounced and accelerated current decay during sustained depolarization, with disproportionately strong suppression of steady-state currents relative to peak currents (Fig. 2). This kinetic profile, together with divergence between concentration-dependent peak and steady-state inhibition, is most consistent with stabilization of an inactivated, nonconducting channel conformation accessed after channel opening.

Further support for this interpretation comes from analysis of voltage-dependent gating. VU426 induced depolarizing shifts in peak  $G$ - $V$  relationships. These effects cannot be readily explained by a simple open-channel block model but instead point to state-dependent modulation of transitions among closed, open, and inactivated states. Notably, low concentrations of VU426 produced modest hyperpolarizing shifts in activation, indicative of facilitation of early gating transitions, whereas higher concentrations favored inhibition (Fig. 3). Similar biphasic behavior has been reported for other ion channel modulators and likely reflects differential engagement of VU426 with distinct conformational states (26, 27). Together, these observations support the classification of VU426 as a gating modifier rather than a pore blocker.

Kv3.1 channels are weakly inactivating (28, 29), yet our data demonstrate that pharmacological modulation can lower the energetic barrier to inactivation, revealing a latent gating pathway that is normally disfavored under physiological conditions. Disease-associated GoF mutations examined here cluster within the S6 transmembrane segment, a pore-lining region that links voltage sensor activation to pore opening (Fig. 4A). Functional analysis confirmed that these mutations shift activation toward hyperpolarized potentials and increase channel open probability, resulting in aberrant current flow at voltages where WT channels are quiescent (Fig. 4, B–E).

Importantly, these same GoF variants exhibit enhanced sensitivity to VU426, with up to fivefold increases in apparent potency relative to wild-type channels (Fig. 5A). Variants positioned deeper within the pore (V432M and V434L) displayed the greatest sensitivity, suggesting that conformational changes in the S6 region may favor formation or stabilization of drug-

accessible inactivated states. Although the precise structural determinants of VU426 binding remain to be defined, these data implicate the S6-pore interface as a key contributor to state-dependent modulation and provide new insight into how disease-associated mutations alter the energetic landscape of Kv3.1 gating.

A major motivation for this study was the lack of pharmacological strategies for suppressing Kv3.1 GoF mutations associated with developmental and epileptic encephalopathies. Whereas most Kv3-targeting efforts have focused on enhancing channel activity to compensate for loss-of-function (30, 31), our data demonstrate that GoF phenotypes are also pharmacologically tractable. VU426 not only inhibits GoF channels with higher potency than wild type but also shifts their voltage dependence of activation in the depolarizing direction, partially normalizing pathological hyperactivity. In several cases, activation thresholds were restored to values indistinguishable from or more depolarized than WT, representing a functional rescue of both increased current amplitude and aberrant gating. These findings are particularly notable in light of recent reports demonstrating preferential inhibition of a disease-associated Kv3.1(V425M) (11) and Kv3.2(V473A) GoF (32) variants in the S6 domain by fluoxetine. Kv3.1 and Kv3.2 channels share substantial sequence homology, including strong conservation within the S6 transmembrane domain where many pathogenic GoF mutations occur, and both channel subtypes are highly expressed in fast-spiking cortical inhibitory interneurons (28, 33). Moreover, Kv3 family subunits may coassemble into heteromeric channels in native neuronal populations (33, 34), suggesting that principles governing state-dependent pharmacologic sensitivity may extend across closely related Kv3 isoforms. Importantly, excessive Kv-mediated potassium conductance in inhibitory interneurons is predicted to suppress neuronal firing and reduce inhibitory tone, paradoxically promoting network hyperexcitability and epilepsy (32, 35–40). This mechanism may help explain why many Kv3-associated developmental and epileptic encephalopathies are refractory to conventional antiseizure medications, which often act by further suppressing neuronal excitability through sodium channel inhibition. In this context, the ability of VU426 to preferentially suppress hyperactive Kv3.1 channels while relatively sparing WT activity suggests a potentially advantageous therapeutic strategy for selectively normalizing pathological excitability in Kv3-associated epilepsies.

The preferential inhibition of GoF mutations likely reflects differences in state occupancy rather than mutation-specific binding. By increasing open probability and favoring activated conformations, GoF mutations increase access to high-affinity, drug-bound inactivated states, resulting in enhanced apparent potency. This mode of functional selectivity—exploiting disease-induced changes in gating behavior—suggests a general strategy for targeting hyperactive ion channel variants without requiring strict molecular selectivity.

Although VU426 lacks subtype selectivity among Kv channels, inhibiting Kv1, Kv2, and Kv3 family members with overlapping potency, this profile is mechanistically informative. The broad Kv activity of VU426 is consistent with engagement of conserved pore-gating architecture shared across voltage-gated potassium channels, likely within or adjacent to the inner pore cavity where S6 helices and the selectivity filter are coupled. We propose that VU426 allosterically stabilizes an

inactivated conformation through this conserved pore-gate coupling, providing a unifying explanation for its state-dependent inhibition across Kv subfamilies. In this context, the apparent preferential impact of VU426 on Kv3.1 GoF variants arises from altered gating dynamics rather than subtype-specific recognition. Similar to classical Kv modulators such as TEA, which have historically provided foundational insights into channel biophysics despite limited selectivity, VU426 serves as a valuable mechanistic probe. Moreover, identification of a chemical scaffold capable of inducing state-dependent inactivation provides a starting point for future medicinal chemistry efforts aimed at improving subtype specificity while preserving the desired gating effects.

Several important questions remain. First, direct kinetic analyses will be required to quantify the effects of VU426 on inactivation rates, recovery, and use dependence, providing a more complete description of its gating mechanism. Second, extending these studies to neurons and intact circuits will be essential to determine how state-dependent Kv3 inhibition influences parvalbumin interneuron firing and network dynamics in physiological and pathological contexts. Finally, medicinal chemistry optimization guided by structure-function relationships in the S6 region may enable the development of more selective next-generation compounds that exploit conserved gating mechanisms while achieving improved therapeutic windows.

In summary, this work establishes HTS-enabled discovery combined with high-content electrophysiology as a powerful strategy for identifying Kv3.1 inhibitors with novel mechanisms of action. By revealing state-dependent inactivation as a tractable mode of Kv channel modulation and demonstrating pharmacological rescue of *KCNK1* GoF variants, these findings open new avenues for targeting Kv channel-mediated hyperexcitability in neurological disease.

## DATA AVAILABILITY

Data will be made available upon reasonable request.

## SUPPLEMENTAL MATERIAL

Supplemental Figs. S1 and S2: <https://doi.org/10.6084/m9.figshare.32257455>.

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## DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

## AUTHOR CONTRIBUTIONS

R.U.C., V.S.J., E.L.D., and J.S.D. conceived and designed research; R.U.C. and V.S.J. performed experiments; R.U.C., V.S.J., and E.L.D. analyzed data; R.U.C., V.S.J., E.L.D., and J.S.D. interpreted results of experiments; R.U.C., V.S.J., E.L.D., and J.S.D. prepared figures; R.U.C., V.S.J., E.L.D., and J.S.D. drafted manuscript; R.U.C., V.S.J., E.L.D., and J.S.D. edited and revised manuscript; R.U.C., V.S.J., E.L.D., and J.S.D. approved final version of manuscript.

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